Chapter 1

Application Overview

1.1 Purpose of the Application

WorkFlow is a mobile application designed for coordinating field teams. It allows managers to create and assign work orders (jobs) to field workers, track their location in real time on a map, and communicate with them through a built-in chat. Workers receive push notifications about new assignments, can update job statuses, and upload photo documentation directly from the field.

The goal of the application is to replace inefficient email or phone-based coordination and give management an instant overview of job progress and team location.

1.2 Technology Stack

Layer	**Technology**
Frontend	React Native (cross-platform iOS / Android)
Backend	Node.js (Express)
Database	PostgreSQL
Real-time	WebSockets (Socket.IO)
Push notifications	Firebase Cloud Messaging (FCM)
Analytics	Firebase Analytics + Firebase Crashlytics
Authentication	JWT + Google SSO (OAuth 2.0)
Maps	React Native Maps (Google Maps)

Table 1.1: WorkFlow technology stack

User Roles

The application defines two distinct user roles: **Manager** and **Worker**. Each role has a dedicated interface and a specific set of permissions. Roles are strictly separate — a user registered as a Manager cannot act as a Worker and vice versa.

Role Overview

Role	Description
Manager	A team leader responsible for creating and managing projects, assigning jobs to workers, and monitoring worker activity.
Worker	A field employee who receives job assignments, updates task progress, uploads professional certificates, and communicates with their manager.

Role Assignment

Roles are assigned at the point of registration and cannot be changed afterwards. A Manager registers directly through the application. A Worker can only join the system by receiving an invite link generated by a Manager — upon signing up via this link, the Worker is automatically assigned the Worker role and connected to the Manager who sent the invitation.

Chapter 2

UATs

2.1 Positive UATs

Table 2.1: UAT 1+: Assign Job to Worker

Test Case ID	UAT-01+
Description	The manager assigns a job to a worker.
Preconditions	• Manager must be logged in • Manager must have at least one project with at least one worker • Manager must be connected to the internet
Test Scenario	1. Manager opens the Project List and taps an existing project. 2. A list of jobs in the project is displayed. 3. Manager taps a job to open the Job Detail screen. 4. Manager taps the **Assign Worker** button. 5. A pop-up window with available workers is shown. 6. Manager selects a worker and confirms with the **Confirm** button. 7. The assigned worker's name appears in the Job Detail screen.
Output Conditions	The worker is successfully assigned to the job and their name appears in the Job Detail screen.
Test Result	PASS / FAIL

Table 2.2: UAT 2+: Manager Creates a New Project

Test Case ID	UAT-02+
Description	The manager creates a new project.
Preconditions	• Manager is logged in • Internet connection is available
Test Scenario	1. The manager opens the application and navigates to the project list. 2. The manager taps *"Add project"*. 3. A pop-up window appears with default project parameters and options to add tasks. 4. The manager sets the project parameters and adds specific tasks and workers. 5. The manager confirms the creation. 6. The new project appears in the project list.
Output Conditions	The project is successfully created and is visible in the project list.
Test Result	PASS / FAIL

2.2 Negative UATs

Table 2.3: UAT 3+: Create New Task

Test Case ID	UAT-03+
Description	The manager creates a new task within an existing project.
Preconditions	• Manager is logged in • Internet connection is available • The project in which the task will be created already exists
Test Scenario	1. The manager opens the application and navigates to the project list. 2. The manager selects the project in which they want to create the task. 3. The manager opens the Job Detail screen. 4. The manager taps *"Add task"*. 5. The manager enters the task name and optionally assigns workers. 6. The manager confirms the creation.
Output Conditions	The task is successfully created and appears in the task list of the project.
Test Result	PASS / FAIL

Table 2.4: UAT 4+: The Manager Inspects Workers on the Map

Test Case ID	UAT-04+
Description	The manager inspects the location of a worker assigned to a project using the map.
Preconditions	• Manager is logged in • Internet connection is available • The project to which the worker is assigned already exists • The inspected worker is online
Test Scenario	1. The manager opens the application and navigates to the project list. 2. The manager selects the project to which the worker is assigned. 3. The manager taps the *"View Map"* button. 4. The system displays the map and the list of workers assigned to the project. 5. The manager taps on the selected worker. 6. The worker's current location is displayed on the map.
Output Conditions	The worker's location is displayed on the map.
Test Result	PASS / FAIL

Table 2.5: UAT 5+: The Worker Sends a Message to the Manager

Test Case ID	UAT-05+
Description	The worker sends a message to the manager.
Preconditions	• Worker is logged in • Internet connection is available • A chat between the worker and manager already exists
Test Scenario	1. The worker opens the application and navigates to the navigation bar at the bottom of the screen. 2. The worker taps the message icon in the navigation bar. 3. The worker locates the chat with the manager in the chat list and taps on it. 4. The worker writes a message and taps the send icon.
Output Conditions	The message is successfully sent and appears in the chat conversation.
Test Result	PASS / FAIL

Table 2.6: UAT 6+: The Worker Uploads a Plumber Certificate

Test Case ID	UAT-06+
Description	The worker uploads a plumber certificate so the manager can see their area of expertise.
Preconditions	• Worker is logged in • Internet connection is available
Test Scenario	1. The worker opens the application and navigates to the navigation bar at the bottom of the screen. 2. The worker taps the profile icon in the navigation bar. 3. The worker taps the *Edit Profile* button. 4. The worker locates the relevant area of expertise and taps *Upload*. 5. The worker selects the certificate file and confirms the choice. 6. The worker taps *Save* to save the profile changes.
Output Conditions	The certificate is uploaded and visible in the worker's profile.
Test Result	PASS / FAIL

Table 2.7: UAT 7+: The Manager Sends an Invitation to a Worker

Test Case ID	UAT-07+
Description	The manager invites a worker to be connected in the app.
Preconditions	• Manager is logged in • Internet connection is available
Test Scenario	1. The manager opens the application and navigates to the Workers section. 2. The manager taps the *Invite Worker* button. 3. A pop-up window appears displaying the invite link. 4. The manager copies the link and sends it to the worker.
Output Conditions	The invite link is successfully generated and displayed to the manager in a pop-up window.
Test Result	PASS / FAIL

Table 2.8: UAT 1-: Manager Tries to Create a Project with a Duplicate Name

Test Case ID	UAT-01-
Description	The system prevents the manager from creating a project with a name that is already in use.
Preconditions	• Manager is logged in • Internet connection is available • A project with the same name already exists
Test Scenario	1. The manager opens the application and navigates to the project list. 2. The manager taps *"Add project"*. 3. A pop-up window appears with default project parameters. 4. The manager enters a name that is already in use. 5. The manager confirms the action. 6. An error message appears below the input field. 7. The manager enters a unique name and confirms successfully.
Output Conditions	The manager is notified of the duplicate name and successfully creates the project with a unique name.
Test Result	PASS / FAIL

Table 2.9: UAT 2-: Worker Tries to Send a Message Without Internet

Test Case ID	UAT-02-
Description	The system prevents the worker from sending a message when there is no internet connection.
Preconditions	• Worker is logged in • Internet connection is unavailable • A chat between the worker and manager already exists
Test Scenario	1. The worker opens the application and navigates to the navigation bar at the bottom of the screen. 2. The worker taps the message icon in the navigation bar. 3. The worker locates the chat with the manager and taps on it. 4. The worker writes a message and taps the send icon. 5. An error message appears informing the worker that the message could not be sent.
Output Conditions	The message is not sent and an error message is displayed to the worker.
Test Result	PASS / FAIL

Table 2.10: UAT 3-: Manager Tries to Create a Task Without a Name

Test Case ID	UAT-03-
Description	The system prevents the manager from creating a task without entering a name.
Preconditions	• Manager is logged in • Internet connection is available • The project in which the task will be created already exists
Test Scenario	1. The manager opens the application and navigates to the project list. 2. The manager selects the project in which they want to create the task. 3. The manager opens the Job Detail screen. 4. The manager taps *"Add task"*. 5. The manager leaves the task name field empty and confirms the creation. 6. An error message appears below the name input field.
Output Conditions	The task is not created and an error message is displayed below the name input field.
Test Result	PASS / FAIL